

PHYS 357: Honours Quantum Physics 1

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Abstract

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1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Mathematical Background

2.1 Taylor expansion

Definition 1 (Taylor series). *A function f that is smooth near a equals the power series*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n, \quad (1)$$

where $f^{(n)}$ is the n th derivative. Expanding about $a = 0$ gives the Maclaurin series.

Result 1 (Series used in this course).

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad (2)$$

together with the binomial theorem $(1+u)^N = \sum_{n=0}^N \binom{N}{n} u^n$ for integer N .

Remark 1. Comparing the three series gives Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$, hence

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (3)$$

These two are used constantly when rotation matrices are written out.

Problem 1 (The exponential as a limit). Show that $\lim_{N \rightarrow \infty} (1 + x/N)^N = e^x$.

Remark 2. This limit is what lets a rotation be written as an exponential. A finite rotation is N infinitesimal ones stacked up,

$$R(\theta \hat{n}) = \lim_{N \rightarrow \infty} \left(\mathbb{1} - \frac{i}{\hbar} \frac{\theta}{N} \hat{n} \cdot \vec{S} \right)^N = e^{-i\theta \hat{n} \cdot \vec{S} / \hbar}. \quad (4)$$

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3 Stern-Gerlach Experiments

3.1 The classical picture

3.1.1 Angular momentum

Definition 2 (Orbital angular momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (5)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr.$$

Definition 3 (Angular momentum of a rigid body). *For a rigid body¹ rotating about a fixed axis this is*

$$L = I_0\omega,$$

with I_0 the moment of inertia and ω the angular velocity.

Remark 3. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

3.1.2 Magnetic moment of a circulating charge

Definition 4 (Magnetic moment). *A magnetic moment is a measure of the net circulation of electric charge. It is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

The magnetic moment of a point particle of mass m and charge q in a circular orbit of radius r and period T is

$$\mu = \frac{IA}{c} = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z,$$

where I is the current and A is the area enclosed by the orbit. A single circulating charge counts as a steady current $I = q/T = qv/2\pi r$ because it passes any given point on the loop once per period.

3.1.3 Gyromagnetic ratio

Definition 5 (Gyromagnetic ratio). *The gyromagnetic ratio γ is the constant of proportionality between the magnetic moment and the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}. \quad (6)$$

For the circulating charge above,

$$\gamma = \frac{q}{2mc}. \quad (7)$$

¹A rigid body is one that does not deform under the action of external forces.

²Here c is in Gaussian units.

Remark 4. *This is a purely classical result, since the charge is treated as a point particle on a definite orbit with a definite radius and speed, and nothing is quantized. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

3.2 Spin

Definition 6 (Spin). *Spin $\vec{S}$ is an intrinsic angular momentum carried by a particle. It is a fixed property of the particle type, like charge or mass, and does not come from any motion through space. A stationary electron alone in empty space still has spin.*

In quantum mechanics the spin $\vec{S}$ replaces the orbital angular momentum $\vec{L}$ of the classical picture.

3.2.1 Intrinsic moment and the g-factor

The magnetic moment is still proportional to the angular momentum, but the gyromagnetic ratio picks up an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \quad \gamma = \frac{qg}{2mc}, \quad (8)$$

where g is the g-factor of the particle.

Definition 7 (g-factor). *The g-factor of a particle³ is the dimensionless number relating its magnetic moment to its spin angular momentum. It measures how much stronger or weaker the moment is than the classical value $\gamma = q/2mc$, which corresponds to $g = 1$.*

3.2.2 Magnetons

Definition 8 (Planck constant). *The Planck constant h is the fundamental constant of quantum mechanics. It sets the scale at which quantization matters, and relates the energy of a photon to its frequency by $E = h\nu$. Its value is*

$$h \approx 6.626 \times 10^{-34} \text{ J s.}$$

It has units of $J \cdot s$, which is the same as angular momentum.

Definition 9 (Reduced Planck constant).

$$\hbar = \frac{h}{2\pi}. \quad (9)$$

It carries units of angular momentum, so $[\hbar] = [L_z] = \frac{kg \cdot m^2}{s}$. Its value is $\hbar = 1.055 \times 10^{-27} \text{ erg s} = 6.582 \times 10^{-16} \text{ eV s}$.

³Its value must be measured or derived from relativistic quantum theory.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $q\hbar/2mc$ then carries the units and sets the natural size of the moment at a given mass scale.

Definition 10 (Bohr magneton). *The Bohr magneton is the natural unit of magnetic moment for an electron,*

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \quad (10)$$

Definition 11 (Nuclear magneton). *The nuclear magneton is the same combination with the proton mass, and is the natural unit for nucleons,*

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \quad (11)$$

Since $m_p/m_e \approx 1836$, we have $\mu_N \approx \mu_B/1836$.

Definition 12 (Nucleons). *The subatomic particles that make up the nucleus of an atom, specifically protons and neutrons.*

The electron moment is then written as

$$\vec{\mu}_e = g_e \mu_B \frac{\vec{S}}{\hbar}, \quad (12)$$

and similarly for the proton and neutron with μ_N in place of μ_B .

3.2.3 Values for the electron, proton and neutron

Particle	Charge q	Mass m	g -factor	Gyromagnetic ratio γ
Electron	$- e $	m_e	$g_e \approx 2$	$g_e \mu_B / \hbar$
Proton	$+ e $	m_p	$g_p \approx 5.59$	$g_p \mu_N / \hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n \mu_N / \hbar$

Table 1: Charges, masses, g -factors and gyromagnetic ratios.

Remark 5. *The classical derivation gives $\mu = (q/2mc)L_z$, which vanishes when $q = 0$. The neutron is neutral yet has a nonzero moment, so its $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.*

Remark 6. *Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.*

3.3 The Stern-Gerlach apparatus

Definition 13 (Stern-Gerlach apparatus). *A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped to produce a strong field gradient along a chosen axis, here z . The gradient deflects each atom by an amount proportional to μ_z , and a screen records where it lands. Since $\vec{\mu} = \gamma\vec{S}$, the position on the screen is a direct measurement of S_z .*

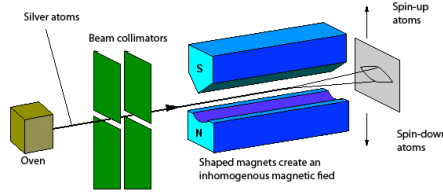


Figure 1: Illustration of the Stern-Gerlach apparatus.

3.3.1 Force in an inhomogeneous field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu} \cdot \vec{B}, \quad (13)$$

so the force on it is

$$\vec{F} = -\vec{\nabla}U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \quad F_z = -\frac{\partial U}{\partial z} = \mu_z \frac{\partial B_z}{\partial z}.$$

Definition 14 (Deflection). *The displacement of an atom on the screen, perpendicular to the beam axis, produced by F_z acting over the length of the magnet. It is proportional to μ_z , so the pattern on the screen is a record of the moments in the beam.*

Remark 7. *A uniform field gives no force, only a torque. Thus, the apparatus needs a field gradient to deflect anything.*

3.3.2 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin. For the electron,

$$\mu_z = -\frac{g_e |e|}{2m_e c} S_z,$$

and therefore

$$F_z = \mu_z \frac{\partial B_z}{\partial z} = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z,$$

where the last equality assumes $\partial B_z / \partial z < 0$.

Result 2.

$$F_z \propto S_z.$$

The deflection of an atom measures the z component of its spin directly.

3.3.3 The classical prediction

Spin is a vector,

$$\vec{S} = (S_x, S_y, S_z)^T, \quad S_z = |\vec{S}| \cos \theta.$$

Remark 8. The spins leaving the oven point in random directions, so θ takes every value in $[0, \pi]$ across the beam and $S_z = |\vec{S}| \cos \theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Since $F_z \propto S_z$, each atom is deflected differently and the screen should show a continuous smear, but it shows two spots.

3.3.4 The observed result

Result 3 (Quantization of spin). The electron spin angular momentum is quantized:

$$S_z = \pm \frac{\hbar}{2}. \quad (14)$$

The beam splits into two spots.

Definition 15 (Spin quantum number). The spin quantum number s labels the magnitude of a particle's spin and is fixed by the particle type. It takes the values $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, and a measurement of S_z then gives one of $2s + 1$ outcomes,

$$S_z = m\hbar, \quad m = -s, -s + 1, \dots, +s. \quad (15)$$

The electron has $s = \frac{1}{2}$, so $S_z = \pm \hbar/2$ and the beam splits in two. A spin-1 particle would split into three beams, $S_z = +\hbar, 0, -\hbar$.

Remark 9. Everything in these notes is the $s = \frac{1}{2}$ case, so the state space is two-dimensional. Note also that S_z is a component, while the magnitude is $|\vec{S}| = \hbar\sqrt{s(s+1)}$, which for the electron is $\hbar\sqrt{3}/2$, not $\hbar/2$.

Definition 16 (Basis state). One of a set of states $\{|j\rangle\}$ that are mutually orthonormal and span the space, so that every state is a unique linear combination of them. For spin- $\frac{1}{2}$ the two outcomes of any one measurement, $\{|\pm\hat{n}\rangle\}$, form a basis.

Definition 17 (Ket notation). A ket $|\cdot\rangle$ labels a quantum state as a vector in a vector space. For electron spin the two measurement outcomes give the basis states

$$|\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \quad |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2}.$$

The space is two-dimensional, so any spin state is a linear combination of the two.

Remark 10. From here on we follow Townsend and write $|+\hat{z}\rangle \equiv |\uparrow\rangle$ and $|-\hat{z}\rangle \equiv |\downarrow\rangle$. The same labels work for any axis: $|\pm\hat{x}\rangle$ are the states with $S_x = \pm\hbar/2$.

3.4 Sequential Stern-Gerlach experiments

Proposition 1. We write SG_z for a device whose gradient points along z , and SG_x for the same device rotated by 90° about the beam axis.

Definition 18 (Filter). Each device sends atoms into two beams. A filter is a device that blocks one of the beams.

Remark 11. A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation.

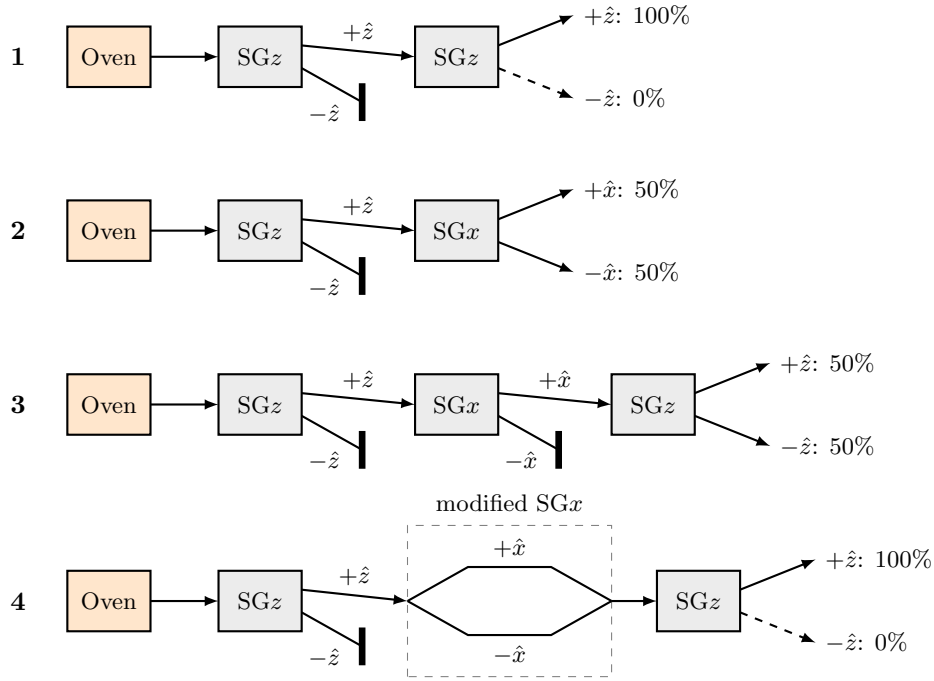


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device. Every oven starts with N_0 atoms.

3.4.1 Experiment 1: SG_z then SG_z

Filter the $+\hat{z}$ beam and send it into a second SG_z . Every atom comes out $+\hat{z}$, none come out $-\hat{z}$. The measurement is repeatable: once an atom is found with $S_z = +\hbar/2$, measuring S_z again gives the same answer. The atom leaves the first device in the state $|+\hat{z}\rangle$. So we end with $N_0/2$ atoms in $|+\hat{z}\rangle$.

3.4.2 Experiment 2: SG_z then SG_x

Filter the $+\hat{z}$ beam and send it into an SG_x. Half the atoms come out $+\hat{x}$ and half come out $-\hat{x}$. Knowing $S_z = +\hbar/2$ tells us nothing about S_x . We end up with $N_0/4$ in $|+\hat{x}\rangle$ and $N_0/4$ in $|-\hat{x}\rangle$.

3.4.3 Experiment 3: SG_z, SG_x, then SG_z

Filter $+\hat{z}$, then filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 into $+\hat{z}$ and $-\hat{z}$ at the end, even though every atom was filtered to $+\hat{z}$ at the start. The S_x measurement wiped out the earlier S_z information. Blocking $+\hat{x}$ instead of $-\hat{x}$ gives the same 50/50 result. We end with $N_0/8$ in $|+\hat{z}\rangle$ and $N_0/8$ in $|-\hat{z}\rangle$.

Result 4. *An atom with a definite value of S_x does not have a definite value of S_z . Measuring S_x destroys what we knew about S_z , so no state has definite values of both at once.*

3.4.4 Experiment 4: SG_z, modified SG_x, then SG_z

Definition 19 (Modified SG device). *An SG device that splits the beam according to S_n and then recombines the two beams, so atoms with $S_n = \pm\hbar/2$ emerge at the same position even though they took two different paths.*

Filter $+\hat{z}$, send it through a modified SG_x with neither path blocked, then measure S_z . Every atom comes out $+\hat{z}$: 100%, not the 50/50 of Experiment 3. All $N_0/2$ atoms end in $|+\hat{z}\rangle$.

Result 5. *The modified SG_x cannot be sending $N_0/4$ through as $|+\hat{x}\rangle$ and $N_0/4$ as $|-\hat{x}\rangle$: Experiment 3 showed each of those alone gives 50/50 at the end. Instead the two paths interfere, as for water or light waves, cancelling the possibility of $-\hat{z}$. No path can be assigned to an atom.*

4 Quantum States as Vectors

4.1 The vector space of spin states

4.1.1 Superposition

Definition 20 (Superposition). *A general spin state is a linear combination of the basis kets,*

$$|\psi\rangle = c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle, \quad c_{\pm} \in \mathbb{C}. \quad (16)$$

The pair $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$ is a basis, and the coefficients $c_{\pm}$ are complex numbers.

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4.1.2 Bras and the inner product

Definition 21 (Bra). *Every ket $|\psi\rangle$ has a matching bra $\langle\psi|$. Putting a bra in front of a ket gives a complex number, the inner product*

$$\langle\phi| |\psi\rangle \equiv \langle\phi|\psi\rangle \in \mathbb{C}.$$

Definition 22 (Orthonormal basis). *The basis kets are orthonormal:*

$$\langle+\hat{z}| +\hat{z}\rangle = \langle-\hat{z}| -\hat{z}\rangle = 1, \quad \langle+\hat{z}| -\hat{z}\rangle = \langle-\hat{z}| +\hat{z}\rangle = 0. \quad (17)$$

4.1.3 Completeness

Definition 23 (Projector). *The operator $P_j = |j\rangle\langle j|$ built from a basis state. Acting on $|\psi\rangle = \sum_k c_k |k\rangle$ it keeps only the $|j\rangle$ part,*

$$P_j |\psi\rangle = c_j |j\rangle,$$

since $\langle j|k\rangle = \delta_{jk}$.

Proposition 2 (Completeness relation). *Since any state can be expanded in the basis $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$, the sum of the projectors onto the basis states is the identity:*

$$|+\hat{z}\rangle\langle+\hat{z}| + |-\hat{z}\rangle\langle-\hat{z}| = \mathbf{1}. \quad (18)$$

Inserting the identity in front of any state gives its expansion directly:

$$|\psi\rangle = (|+\hat{z}\rangle\langle+\hat{z}| + |-\hat{z}\rangle\langle-\hat{z}|) |\psi\rangle = \langle+\hat{z}|\psi\rangle |+\hat{z}\rangle + \langle-\hat{z}|\psi\rangle |-\hat{z}\rangle,$$

so the coefficients $c_{\pm}$ of the superposition are the inner products $\langle\pm\hat{z}|\psi\rangle$.

4.2 Extracting probabilities

4.2.1 Probability amplitudes

Proposition 3. *To pull out c_+ , act on $|\psi\rangle$ with $\langle+\hat{z}|$:*

$$\langle+\hat{z}|\psi\rangle = c_+ \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + c_- \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = c_+.$$

The same steps with $\langle-\hat{z}|$ give c_- . So

$$c_{\pm} = \langle\pm\hat{z}|\psi\rangle. \quad (19)$$

Definition 24 (Probability amplitude). *The coefficient $c_{\pm} = \langle\pm\hat{z}|\psi\rangle$ is the probability amplitude for measuring $S_z = \pm\hbar/2$. The probability itself is the modulus squared,*

$$P(S_z = \pm\frac{\hbar}{2}) = |c_{\pm}|^2 = |\langle\pm\hat{z}|\psi\rangle|^2. \quad (20)$$

Principle 1 (Born rule). *The probability that a system in the state $|\psi\rangle$ is found in the state $|\phi\rangle$ is*

$$P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 \in [0, 1]. \quad (21)$$

Remark 12. *Why is the Born rule true? There is no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."*

4.2.2 Normalization

Proposition 4. *A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, so*

$$P(\psi|\psi) = |\langle\psi|\psi\rangle|^2 = 1.$$

To find what a bra equals to, write it with unknown coefficients, $\langle\psi| = c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|$. Then

$$\langle\psi|\psi\rangle = (c'_+ \langle+\hat{z}| + c'_- \langle-\hat{z}|) (c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle) = c'_+ c_+ + c'_- c_-,$$

since the cross terms vanish by orthonormality. The two outcomes $S_z = \pm\hbar/2$ must have probabilities $|c_+|^2$ and $|c_-|^2$ that add to 1. For $\langle\psi|\psi\rangle$ to be that sum, each product $c'_\pm c_\pm$ must be real and equal to $|c_\pm|^2$. This forces $c'_\pm = c_\pm^*$.

Result 6. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle\psi| = c_+^* \langle+\hat{z}| + c_-^* \langle-\hat{z}|. \quad (22)$$

In particular $\langle\psi|\pm\hat{z}\rangle = c_\pm^* = \langle\pm\hat{z}|\psi\rangle^*$. More generally, $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$.

Definition 25 (Normalization condition).

$$\langle\psi|\psi\rangle = \langle\psi|+\hat{z}\rangle \langle+\hat{z}|\psi\rangle + \langle\psi|-\hat{z}\rangle \langle-\hat{z}|\psi\rangle = |c_+|^2 + |c_-|^2 = 1. \quad (23)$$

In words, $P(S_z = +\hbar/2) + P(S_z = -\hbar/2) = 1$: the atom is always found in one of the two beams.

Example. Take

$$|\psi\rangle = \frac{1}{\sqrt{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle.$$

The amplitude for $S_z = +\hbar/2$ is

$$\langle+\hat{z}|\psi\rangle = \frac{1}{\sqrt{3}} \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + i\sqrt{\frac{2}{3}} \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = \frac{1}{\sqrt{3}},$$

so $P(S_z = +\hbar/2) = 1/3$. In the same way $\langle-\hat{z}|\psi\rangle = i\sqrt{2/3}$, so $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. The two add to 1, so the state is normalized.

4.2.3 Global phase

Proposition 5. *Take two kets that differ only by an overall phase, $|\psi\rangle = e^{i\alpha} |\psi'\rangle$ with $\alpha \in \mathbb{R}$. For any state $|\phi\rangle$,*

$$P(\phi|\psi) = |\langle\phi|\psi\rangle|^2 = |e^{i\alpha} \langle\phi|\psi'\rangle|^2 = |\langle\phi|\psi'\rangle|^2 = P(\phi|\psi'),$$

since $|e^{i\alpha}| = 1$.

Result 7. $|\psi\rangle$ and $e^{i\alpha} |\psi\rangle$ give the same probabilities for every possible measurement, so they describe the same physical state. An overall (global) phase has no physical meaning and can be dropped.

Remark 13. A relative phase is different: it sits between the two terms of a superposition, so it cannot be pulled out front and dropped. It has physical consequences, as the next subsection shows.

4.3 Fixing the basis states

4.3.1 The state $|+\hat{x}\rangle$

In Experiment 3, atoms in $|+\hat{x}\rangle$ enter an SG z and split 50/50:

$$P(S_z = \pm \frac{\hbar}{2}) = |\langle \pm \hat{z} | +\hat{x} \rangle|^2 = \frac{1}{2}.$$

So both amplitudes have modulus $1/\sqrt{2}$, and each can carry its own phase:

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} e^{i\delta_+} |+\hat{z}\rangle + \frac{1}{\sqrt{2}} e^{i\delta_-} |-\hat{z}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle), \quad \delta = \delta_- - \delta_+,$$

with $\delta_{\pm} \in \mathbb{R}$. Dropping the global phase $e^{i\delta_+}$ leaves

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle).$$

Remark 14. *The relative phase δ survives, and Experiment 3 alone does not fix it. It is pinned down by combining more measurements (for example along y) with a phase convention.*

4.3.2 Experiment 5: measuring S_y

Definition 26 (Cyclic permutation). *A cyclic permutation of the axes relabels them as $x \rightarrow y$, $y \rightarrow z$, $z \rightarrow x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes under it.*

Let Experiment 5 be a cyclic permutation of Experiment 3, SG x becomes SG z and SG z becomes SG y . Now the process is as follows: filter $+\hat{z}$, then measure S_y .

Proposition 6. *By symmetry it must split 50/50 as well, so every cross-axis overlap has the same modulus:*

$$|\langle +\hat{x} | +\hat{z} \rangle|^2 = |\langle +\hat{y} | +\hat{z} \rangle|^2 = |\langle -\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2}.$$

The same argument as for $|+\hat{x}\rangle$ then gives

$$|+\hat{y}\rangle = \frac{e^{i\gamma_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\gamma} |-\hat{z}\rangle), \quad \gamma = \gamma_- - \gamma_+.$$

4.3.3 The state $|+\hat{y}\rangle$ and the phase convention

Proposition 7. *Dropping the global phases, $\gamma_+ = \delta_+ = 0$, we obtain equations for x and y in terms of z :*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\delta} |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm e^{i\gamma} |-\hat{z}\rangle).$$

So using orthonormality of the $\hat{z}$ basis,

$$\langle +\hat{x} | +\hat{y} \rangle = \frac{1}{2} (1 + e^{i(\gamma-\delta)}), \quad |\langle +\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2} (1 + \cos(\gamma - \delta)).$$

Requiring this to equal 1/2 forces $\cos(\gamma - \delta) = 0$, that is $\gamma - \delta = \pm\pi/2$. The remaining freedom is convention.

Result 8 (Phase convention). *Take $\delta = 0$ and $\gamma = +\pi/2$. Then*

$$|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm |-\hat{z}\rangle), \quad |\pm\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle \pm i |-\hat{z}\rangle). \quad (24)$$

The bras are then

$$\langle \pm\hat{x} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \pm \langle -\hat{z} |), \quad \langle \pm\hat{y} | = \frac{1}{\sqrt{2}} (\langle +\hat{z} | \mp i \langle -\hat{z} |). \quad (25)$$

Remark 15. *Complex coefficients are required, not just allowed. With real coefficients the relative phases could only be 0 or π , so $\gamma - \delta = \pm\pi/2$ would be impossible.*

Example. For $|\psi\rangle = \sqrt{\frac{1}{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle$, compute $\langle S_y \rangle$.
The amplitudes are

$$\begin{aligned} \langle +\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle +\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle +\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (-i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + \sqrt{\frac{2}{3}} \right), \\ \langle -\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle -\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle -\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} - \sqrt{\frac{2}{3}} \right). \end{aligned}$$

Computing the probabilities

$$P(S_y = +\frac{\hbar}{2}) = |\langle +\hat{y} | \psi \rangle|^2 = \frac{1}{2} + \frac{\sqrt{2}}{3}, \quad P(S_y = -\frac{\hbar}{2}) = |\langle -\hat{y} | \psi \rangle|^2 = \frac{1}{2} - \frac{\sqrt{2}}{3},$$

which add to 1 as they must. Then

$$\langle S_y \rangle = \frac{\hbar}{2} \left[\left(\frac{1}{2} + \frac{\sqrt{2}}{3} \right) - \left(\frac{1}{2} - \frac{\sqrt{2}}{3} \right) \right] = \frac{\sqrt{2}}{3} \hbar.$$

The factor i that did nothing for S_z matters here: it is a relative phase, and it is exactly what tilts the spin toward $+\hat{y}$.

4.4 Statistics of measurements

4.4.1 Expectation value

Definition 27 (Expectation value). *The expectation value of S_z in the state $|\psi\rangle$ is the average outcome weighted by the probabilities,*

$$\langle S_z \rangle = \left(\frac{\hbar}{2} \right) |\langle +\hat{z} | \psi \rangle|^2 + \left(-\frac{\hbar}{2} \right) |\langle -\hat{z} | \psi \rangle|^2. \quad (26)$$

Remark 16. For N measurements on identically prepared atoms, with N_+ giving $+\hbar/2$ and N_- giving $-\hbar/2$, the measured average is

$$\langle S_z \rangle_N = \frac{\hbar}{2} \frac{N_+ - N_-}{N},$$

which tends to $\langle S_z \rangle$ as $N \rightarrow \infty$.

Remark 17 (Shot noise). For a finite number N of measurements, the sample average deviates from the expectation value by

$$|\langle S_z \rangle_N - \langle S_z \rangle| \sim \frac{\Delta S_z}{\sqrt{N}}.$$

This spread is intrinsic, not a flaw of the apparatus: it appears whenever a state with a definite value of one observable (e.g. S_x for $|\hat{x}\rangle$) is measured for an incompatible observable (e.g. S_z).

4.4.2 Uncertainty

Definition 28 (Uncertainty). The uncertainty ΔS_z is the standard deviation of the measurement outcomes,

$$(\Delta S_z)^2 = \langle (S_z - \langle S_z \rangle)^2 \rangle = \langle S_z^2 \rangle - \langle S_z \rangle^2, \quad (27)$$

where

$$\langle S_z^2 \rangle = \left(\frac{\hbar}{2}\right)^2 |\langle +\hat{z}|\psi \rangle|^2 + \left(-\frac{\hbar}{2}\right)^2 |\langle -\hat{z}|\psi \rangle|^2. \quad (28)$$

Example. For $|\hat{x}\rangle$, both probabilities are $1/2$, so

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{2} - \frac{\hbar}{2} \cdot \frac{1}{2} = 0, \quad \langle S_z^2 \rangle = \left(\frac{\hbar}{2}\right)^2 \cdot \frac{1}{2} + \left(-\frac{\hbar}{2}\right)^2 \cdot \frac{1}{2} = \frac{\hbar^2}{4},$$

and therefore $\Delta S_z = \hbar/2$. The average is zero but every single measurement gives $\pm\hbar/2$, so the spread is maximal.

Example. For $|\psi\rangle = \sqrt{\frac{1}{3}}|\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, the probabilities are $1/3$ and $2/3$, so

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{3} - \frac{\hbar}{2} \cdot \frac{2}{3} = -\frac{\hbar}{6}, \quad \langle S_z^2 \rangle = \frac{\hbar^2}{4} \left(\frac{1}{3} + \frac{2}{3}\right) = \frac{\hbar^2}{4}.$$

Then

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{9\hbar^2 - \hbar^2}{36} = \frac{2}{9}\hbar^2, \quad \Delta S_z = \frac{\sqrt{2}}{3}\hbar.$$

Remark 18. For spin- $\frac{1}{2}$, $\langle S_z^2 \rangle$ is always equal to $\frac{\hbar^2}{4}$, since both outcomes square to the same number and the probabilities sum to 1.

4.5 Generalization to n -level systems

4.5.1 Orthonormality and completeness

A spin- $\frac{1}{2}$ particle is a two-level system, and each of $\{|\pm\hat{x}\rangle\}$, $\{|\pm\hat{y}\rangle\}$, $\{|\pm\hat{z}\rangle\}$ is a basis. An n -level system has a basis $\{|j\rangle\}$, $j = 1, \dots, n$.

Remark 19. The index j labels the outcomes of one chosen observable, not the coordinate axes. Picking S_x , S_y or S_z picks the basis; within that basis $j = 1, 2$ labels its two states, for example $|a_1\rangle = |+\hat{x}\rangle$ and $|a_2\rangle = |-\hat{x}\rangle$. A spin-1 particle has $n = 3$ outcomes, and an n -level atom has $j = 1, \dots, n$.

Definition 29 (Orthonormal and complete basis). The basis is orthonormal if

$$\langle i | j \rangle = \delta_{ij},$$

and complete if every state can be expanded in it,

$$|\psi\rangle = \sum_j c_j |j\rangle, \quad c_j = \langle j | \psi \rangle, \quad \langle \psi | = \sum_j c_j^* \langle j|. \quad (29)$$

Completeness is equivalent to

$$\mathbb{1} = \sum_{j=1}^n |j\rangle \langle j|. \quad (30)$$

Example. For $n = 2$, $\mathbb{1} = |+\hat{n}\rangle \langle +\hat{n}| + |-\hat{n}\rangle \langle -\hat{n}|$ for any $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$, so

$$|\psi\rangle = \langle +\hat{x} | \psi \rangle |+\hat{x}\rangle + \langle -\hat{x} | \psi \rangle |-\hat{x}\rangle = \langle +\hat{y} | \psi \rangle |+\hat{y}\rangle + \langle -\hat{y} | \psi \rangle |-\hat{y}\rangle.$$

4.5.2 Observables and the Born rule

Definition 30 (Observable). An observable A has outcomes a_j with states $|a_j\rangle$ forming a complete orthonormal set. For spin- $\frac{1}{2}$, A could be S_x , S_y or S_z , with outcomes $\pm\hbar/2$. By the Born rule,

$$P(A = a_j) = |\langle a_j | \psi \rangle|^2. \quad (31)$$

Proposition 8. Normalization $\langle \psi | \psi \rangle = 1$ is the statement that the probabilities sum to 1.

Proof.

$$\langle \psi | \psi \rangle = \left(\sum_i c_i^* \langle a_i | \right) \left(\sum_j c_j |a_j\rangle \right) = \sum_{i,j} c_i^* c_j \delta_{ij} = \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.$$

Equivalently, inserting the identity, $\sum_j \langle \psi | a_j \rangle \langle a_j | \psi \rangle = \langle \psi | \left(\sum_j |a_j\rangle \langle a_j| \right) | \psi \rangle = \langle \psi | \psi \rangle$. $\square$

4.5.3 Expectation value as an operator

Proposition 9.

$$\langle A \rangle = \sum_j a_j |\langle a_j | \psi \rangle|^2 = \sum_j \langle \psi | a_j \rangle a_j \langle a_j | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle, \quad \hat{A} = \sum_j a_j |a_j\rangle \langle a_j|. \quad (32)$$

Definition 31 (Uncertainty of an observable).

$$(\Delta A)^2 = \langle (A - \langle A \rangle)^2 \rangle = \langle A^2 \rangle - \langle A \rangle^2, \quad \langle A^2 \rangle = \langle \psi | \hat{A}^2 | \psi \rangle = \sum_j a_j^2 |c_j|^2. \quad (33)$$

5 Rotation of Basis States and Matrix Mechanics

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5.1 States as column vectors

Definition 32 (Representation of a state). *In the S_z basis, a ket is represented by the column of its components and a bra by the conjugate row, the adjoint:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \quad \langle \psi | \xrightarrow{S_z} \psi^\dagger = (c_+^* \quad c_-^*), \quad (34)$$

with $c_\pm = \langle \pm \hat{z} | \psi \rangle$. Then

$$\langle \psi | \psi \rangle = \psi^\dagger \psi = |c_+|^2 + |c_-|^2. \quad (35)$$

Remark 20. Write $|\psi\rangle \xrightarrow{S_z} \psi$, not $|\psi\rangle = \psi$. The ket is basis-independent; the column depends on the basis.

Example.

$$|\pm \hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad |\pm \hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Problem 2 (Other bases). *In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | \psi \rangle \\ \langle -\hat{x} | \psi \rangle \end{pmatrix}$, and similarly for S_y . Write the column representing $|+\hat{z}\rangle$ in the S_z , S_x and S_y bases.*

Proposition 10 (Inner product as a matrix product). *If $|\phi\rangle \xrightarrow{S_z} \phi$ and $|\psi\rangle \xrightarrow{S_z} \psi$, then the inner product is a row times a column,*

$$\langle \phi | \psi \rangle = \phi^\dagger \psi, \quad (36)$$

and the Born rule reads $P(\phi | \psi) = |\phi^\dagger \psi|^2$.

Remark 21. If the state is given in another basis, first replace each basis ket by its S_z column and add. For $|\psi\rangle = a|+\hat{y}\rangle + b|-\hat{y}\rangle$,

$$\psi \xrightarrow{S_z} \frac{a}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} + \frac{b}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a+b \\ i(a-b) \end{pmatrix},$$

and $\psi^\dagger \psi = 1$ is worth checking before going further.

Problem 3 (A probability by matrix mechanics). For $|\psi\rangle = \sqrt{\frac{1}{5}}|+\hat{z}\rangle + i\sqrt{\frac{4}{5}}|-\hat{z}\rangle$, find $P(S_x = +\hbar/2)$ using matrix mechanics.

5.2 Operators as matrices

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Definition 33 (Projection operator). $P_{\pm\hat{n}} = |\pm\hat{n}\rangle\langle\pm\hat{n}|$, so that $\mathbb{1} = P_{+\hat{n}} + P_{-\hat{n}}$ and $P_{+\hat{z}}|\psi\rangle = c_+|+\hat{z}\rangle$.

Definition 34 (Matrix of an operator). In a basis $\{|j\rangle\}$, an operator A is represented by the matrix

$$A_{ij} = \langle i|A|j\rangle. \quad (37)$$

Proposition 11 (Operator acting on a state). Acting with an operator on a ket becomes a matrix times a column vector,

$$A|\psi\rangle \rightarrow A\psi.$$

Proof. The k th entry of the column representing $A|\psi\rangle$ is its overlap with the k th basis state, $\langle k|A|\psi\rangle$. Insert $\mathbb{1} = \sum_j |j\rangle\langle j|$ between A and $|\psi\rangle$:

$$\langle k|A|\psi\rangle = \sum_j \underbrace{\langle k|A|j\rangle}_{A_{kj}} \underbrace{\langle j|\psi\rangle}_{c_j} = \sum_j A_{kj} c_j.$$

This is exactly row k of the matrix A multiplied by the column $\psi = (c_1, c_2, \dots)^T$, which is the rule for matrix-times-vector multiplication. $\square$

Example.

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix} \langle +\hat{z}|P_{+\hat{z}}|+\hat{z}\rangle & \langle +\hat{z}|P_{+\hat{z}}|-\hat{z}\rangle \\ \langle -\hat{z}|P_{+\hat{z}}|+\hat{z}\rangle & \langle -\hat{z}|P_{+\hat{z}}|-\hat{z}\rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} c_+ \\ c_- \end{pmatrix} = \begin{pmatrix} c_+ \\ 0 \end{pmatrix}.$$

Proposition 12 (Properties of projectors). A projector satisfies $P_{\hat{n}}^2 = P_{\hat{n}}$, since $(|\hat{n}\rangle\langle\hat{n}|)(|\hat{n}\rangle\langle\hat{n}|) = |\hat{n}\rangle\langle\hat{n}|\hat{n}\rangle\langle\hat{n}| = |\hat{n}\rangle\langle\hat{n}|$. In the S_z basis,

$$P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbb{1} = P_{+\hat{z}} + P_{-\hat{z}} \xrightarrow{S_z} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Definition 35 (Adjoint). Writing $A = \mathbb{1}A\mathbb{1} = \sum_{i,j} A_{ij}|i\rangle\langle j|$, the adjoint is

$$A^\dagger = \sum_{i,j} A_{ij}^* |j\rangle\langle i|, \quad (A^\dagger)_{ij} = A_{ji}^*. \quad (38)$$

Proposition 13 (Bra of a transformed state). *If $|\psi'\rangle = A|\psi\rangle$, then $\langle\psi'| = \langle\psi|A^\dagger$.*

Proposition 14 (Adjoint of a product).

$$(AB)^\dagger = B^\dagger A^\dagger. \quad (39)$$

Remark 22. *The order reverses, and $A^\dagger B^\dagger$ is a different matrix in general. For real entries the adjoint is just the transpose.*

5.3 The spin operators

Each spin component is built from the projectors onto its own basis, as the operator form of the expectation value requires:

$$\hat{S}_n = \frac{\hbar}{2} (|+\hat{n}\rangle \langle +\hat{n}| - |-\hat{n}\rangle \langle -\hat{n}|), \quad \hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}. \quad (40)$$

Its matrix in the S_z basis is $(\hat{S}_n)_{ij} = \langle i | \hat{S}_n | j \rangle$ with $i, j \in \{+\hat{z}, -\hat{z}\}$. Each entry is a difference of two products, one overlap from the bra side and one from the ket side, and the bra side is conjugated.

Problem 4 (The matrix of $\hat{S}_z$). *Find the matrix representation of $\hat{S}_z$ in the S_z basis.*

Problem 5 (The matrix of $\hat{S}_x$). *Find the matrix representation of $\hat{S}_x$ in the S_z basis.*

Problem 6 (The matrix of $\hat{S}_y$). *Find the matrix representation of $\hat{S}_y$ in the S_z basis.*

Result 9 (Pauli matrices).

$$\hat{S}_j = \frac{\hbar}{2} \sigma_j, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (41)$$

all in the S_z basis. Each is its own adjoint, and each squares to $\mathbb{1}$, so $\hat{S}_j^2 = \frac{\hbar^2}{4} \mathbb{1}$ for every j .

Remark 23 (Shortcut). *Instead of four separate brackets, write each projector as an outer product of a column with its conjugate row, for example*

$$|+\hat{x}\rangle \langle +\hat{x}| \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad |-\hat{x}\rangle \langle -\hat{x}| \xrightarrow{S_z} \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

then subtract and multiply by $\hbar/2$.

Problem 7 (Expectation value and uncertainty by matrix mechanics). *For $|\psi\rangle = \sqrt{\frac{1}{5}}|+\hat{y}\rangle + i\sqrt{\frac{4}{5}}|-\hat{y}\rangle$, find $\langle S_z \rangle$ and ΔS_z using matrix mechanics.*

5.4 The rotation operator

Definition 36 (Rotation operator). $R(\theta\hat{n})$ rotates a state by θ about $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})|\psi\rangle$. For example, $R(\frac{\pi}{2}\hat{y})$ takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ (up to phase), so

$$R\left(\frac{\pi}{2}\hat{y}\right)(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+|+\hat{x}\rangle + c_-|-\hat{x}\rangle.$$

Remark 24. $R(\theta\hat{n})$ is a linear operator: $R(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+R|+\hat{z}\rangle + c_-R|-\hat{z}\rangle$, which is what the equation above uses.

Remark 25. The bra of $|\psi'\rangle$ is $\langle\psi|R^\dagger$, not $\langle\psi|R$.

Theorem 1. Rotation operators are unitary:

$$R^\dagger R = RR^\dagger = \mathbb{1}. \quad (42)$$

Proof. A rotation must preserve normalization: $\langle\psi'|\psi'\rangle = \langle\psi|R^\dagger R|\psi\rangle = 1$ for every $|\psi\rangle$ and every $R(\theta\hat{n})$, which forces $R^\dagger R = \mathbb{1}$. $\square$

Definition 37 (Unitary Operator). Unitary operators map physical (normalized) states to physical states. If a unitary U satisfies $U|\psi\rangle = c|\psi\rangle$, then $|c| = 1$, so $c = e^{i\varphi}$ is just a global phase and $U|\psi\rangle$ describes the same physical state as $|\psi\rangle$.

5.4.1 Rotations about $\hat{z}$

Result 10. A rotation about $\hat{z}$ cannot change S_z , so it leaves the S_z basis states where they are and can only multiply them by a phase:

$$R(\phi\hat{z})|\pm\hat{z}\rangle = e^{\mp i\phi/2}|\pm\hat{z}\rangle.$$

The exponent follows from $R = e^{-i\phi\hat{S}_z/\hbar}$ acting on a state with $S_z = \pm\hbar/2$.

Remark 26. A full turn, $\phi = 2\pi$, gives $R = -\mathbb{1}$ rather than $\mathbb{1}$: a spin- $\frac{1}{2}$ state needs 4π to come back to itself. The sign is a global phase, so it is unobservable on its own.

Remark 27. In the S_z basis the matrix is diagonal,

$$R(\phi\hat{z}) \xrightarrow{S_z} \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix},$$

but in any other basis it is not, and it has to be built entry by entry from $R_{ij} = \langle i|R|j\rangle$.

Problem 8 ($R(\phi\hat{z})$ in the S_x basis). Find the matrix representation of $R(\phi\hat{z})$ in the S_x basis.

Remark 28. Repeating the calculation in the S_y basis, where the bras carry $\mp i$, gives the same matrix. Both $\hat{x}$ and $\hat{y}$ are perpendicular to the rotation axis, so the rotation looks identical from either one.

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5.4.2 Rotation as a change of basis

Rotating the S_z basis by 90° about $\hat{y}$ turns it into the S_x basis, so we require

$$R\left(\frac{\pi}{2}\hat{y}\right)|\pm\hat{z}\rangle = |\pm\hat{x}\rangle.$$

Multiplying from the left by $\langle\pm\hat{z}|$ gives the matrix elements of $R\left(\frac{\pi}{2}\hat{y}\right)$ in the S_z basis:

$$R\left(\frac{\pi}{2}\hat{y}\right) \xrightarrow{S_z} \begin{pmatrix} \langle+\hat{z}|R|+\hat{z}\rangle & \langle+\hat{z}|R|-\hat{z}\rangle \\ \langle-\hat{z}|R|+\hat{z}\rangle & \langle-\hat{z}|R|-\hat{z}\rangle \end{pmatrix} = \begin{pmatrix} \langle+\hat{z}|+\hat{x}\rangle & \langle+\hat{z}|-\hat{x}\rangle \\ \langle-\hat{z}|+\hat{x}\rangle & \langle-\hat{z}|-\hat{x}\rangle \end{pmatrix}.$$

From $|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}(|+\hat{z}\rangle \pm |-\hat{z}\rangle)$ and orthonormality, $\langle+\hat{z}|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}}$ and $\langle-\hat{z}|\pm\hat{x}\rangle = \pm\frac{1}{\sqrt{2}}$.

Result 11.

$$R\left(\frac{\pi}{2}\hat{y}\right) \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Remark 29. The columns of the matrix are the target kets written as S_z columns: the first column is $|\hat{x}\rangle$ and the second is $|\hat{-x}\rangle$. The same shortcut gives any other change-of-basis rotation, for example the one taking $|\pm\hat{z}\rangle$ to $|\pm\hat{y}\rangle$ has columns $\frac{1}{\sqrt{2}}(1, \pm i)^T$.

Now take a state given in the S_x basis,

$$|\psi\rangle = \langle+\hat{x}|\psi\rangle|\hat{x}\rangle + \langle-\hat{x}|\psi\rangle|\hat{-x}\rangle \xrightarrow{S_x} \psi^x = \begin{pmatrix} \langle+\hat{x}|\psi\rangle \\ \langle-\hat{x}|\psi\rangle \end{pmatrix}.$$

To find its S_z column, insert $\mathbf{1} = |\hat{x}\rangle\langle+\hat{x}| + |\hat{-x}\rangle\langle-\hat{x}|$ into each S_z amplitude:

$$\langle\pm\hat{z}|\psi\rangle = \langle\pm\hat{z}|+\hat{x}\rangle\langle+\hat{x}|\psi\rangle + \langle\pm\hat{z}|-\hat{x}\rangle\langle-\hat{x}|\psi\rangle.$$

In matrix form this reads

$$\psi^z = \begin{pmatrix} \langle+\hat{z}|\psi\rangle \\ \langle-\hat{z}|\psi\rangle \end{pmatrix} = \begin{pmatrix} \langle+\hat{z}|+\hat{x}\rangle & \langle+\hat{z}|-\hat{x}\rangle \\ \langle-\hat{z}|+\hat{x}\rangle & \langle-\hat{z}|-\hat{x}\rangle \end{pmatrix} \begin{pmatrix} \langle+\hat{x}|\psi\rangle \\ \langle-\hat{x}|\psi\rangle \end{pmatrix},$$

and the 2×2 matrix is exactly the one found above.

Result 12 (Change of basis).

$$\psi^z = R\left(\frac{\pi}{2}\hat{y}\right)\psi^x, \quad \psi^x = R^\dagger\left(\frac{\pi}{2}\hat{y}\right)\psi^z, \quad (43)$$

where the second follows from unitarity.

Result 13 (Expectation Value). With the state in the S_z basis, expectation values follow from

$$\langle S_z \rangle = \psi^{z\dagger} \hat{S}_z \psi^z. \quad (44)$$

Problem 9 (Expectation value from the S_x basis). For $|\psi\rangle = \frac{1}{\sqrt{3}}|\hat{x}\rangle + i\sqrt{\frac{2}{3}}|\hat{-x}\rangle$, find $\langle S_z \rangle$ using matrix mechanics.

6 Change of Basis in n -Level Systems

6.1 Two bases for the same space

Take an n -level system with two orthonormal bases,

$$\begin{aligned} \{|j\rangle : j = 1, \dots, n\}, \quad & \{|\alpha\rangle : \alpha = 1, \dots, n\}, \\ \langle i|j\rangle = \delta_{ij}, \quad & \langle \alpha|\beta\rangle = \delta_{\alpha\beta}. \end{aligned}$$

Remark 30. The cross overlaps are not Kronecker deltas: $\langle j|\alpha\rangle \neq \delta_{j\alpha}$ in general, just as $\langle +\hat{x}|\hat{z}\rangle = \frac{1}{\sqrt{2}}$ is neither 0 nor 1.

Both bases are complete, so the identity can be inserted either way:

$$|\psi\rangle = \mathbb{1} |\psi\rangle = \sum_{j=1}^n |j\rangle \langle j|\psi\rangle = \sum_{\alpha=1}^n |\alpha\rangle \langle \alpha|\psi\rangle. \quad (45)$$

The same ket then has two column representations,

$$|\psi\rangle \xrightarrow{|j\rangle} \psi^j = \begin{pmatrix} \langle 1|\psi\rangle \\ \langle 2|\psi\rangle \\ \vdots \\ \langle n|\psi\rangle \end{pmatrix}, \quad |\psi\rangle \xrightarrow{|\alpha\rangle} \psi^\alpha = \begin{pmatrix} \langle \alpha=1|\psi\rangle \\ \langle \alpha=2|\psi\rangle \\ \vdots \\ \langle \alpha=n|\psi\rangle \end{pmatrix}. \quad (46)$$

6.2 The transformation matrix

Inserting $\mathbb{1} = \sum_{\alpha} |\alpha\rangle \langle \alpha|$ into each $|j\rangle$ amplitude,

$$\langle j|\psi\rangle = \sum_{\alpha=1}^n \langle j|\alpha\rangle \langle \alpha|\psi\rangle,$$

which is a matrix times a column.

Result 14 (Change of basis).

$$\psi^j = U \psi^\alpha, \quad U_{j\alpha} = \langle j|\alpha\rangle. \quad (47)$$

The columns of U are the new basis kets $|\alpha\rangle$ written in the $|j\rangle$ basis.

Theorem 2. U is unitary: $U^\dagger U = U U^\dagger = \mathbb{1}$.

Proof. Using $(U^\dagger)_{\alpha j} = U_{j\alpha}^* = \langle \alpha|j\rangle$ and completeness of $\{|j\rangle\}$,

$$(U^\dagger U)_{\alpha\beta} = \sum_j \langle \alpha|j\rangle \langle j|\beta\rangle = \langle \alpha|\beta\rangle = \delta_{\alpha\beta}.$$

The same argument with completeness of $\{|\alpha\rangle\}$ gives $U U^\dagger = \mathbb{1}$. $\square$

Remark 31. Normalization is basis-independent, as it must be:

$$\langle \psi|\psi\rangle = \psi^{j\dagger} \psi^j = \psi^{\alpha\dagger} U^\dagger U \psi^\alpha = \psi^{\alpha\dagger} \psi^\alpha = 1.$$

Example. With $|j\rangle = |\pm\hat{z}\rangle$ and $|\alpha\rangle = |\pm\hat{x}\rangle$, $U_{j\alpha} = \langle \pm\hat{z}|\pm\hat{x}\rangle$ is exactly $R(\frac{\pi}{2}\hat{y})$ from the previous section, and $\psi^j = U \psi^\alpha$ is $\psi^z = R \psi^x$.

6.3 Eigenstates and eigenvalues

Definition 38 (Eigenstate). A state $|a_i\rangle$ is an eigenstate of the operator $\hat{A}$ with eigenvalue a_i if

$$\hat{A}|a_i\rangle = a_i|a_i\rangle. \quad (48)$$

In matrix form this is the familiar $Mv = \lambda v$, with v the eigenvector.

Remark 32. A general operator maps a state to a different state, $\hat{A}|\psi\rangle = c|\phi\rangle$. Eigenstates are the special states that come back to themselves up to a number.

Example.

$$\hat{S}_z|\pm\hat{z}\rangle = \pm\frac{\hbar}{2}|\pm\hat{z}\rangle,$$

so $|\pm\hat{z}\rangle$ are the eigenstates of $\hat{S}_z$ and $\pm\hbar/2$ its eigenvalues, i.e. the measurement outcomes.

Proposition 15 (An operator is diagonal in its own eigenbasis).

$$A_{ij} = \langle a_i|\hat{A}|a_j\rangle = a_j\langle a_i|a_j\rangle = a_j\delta_{ij}, \quad (49)$$

so

$$\hat{A} \xrightarrow{|a_j\rangle} D = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{pmatrix}. \quad (50)$$

In particular $\langle a_i|\hat{A}|a_i\rangle = a_i$.

Example. $\hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

6.4 Operators in another basis

Let $|j\rangle = |a_j\rangle$ be the eigenbasis of $\hat{A}$, so $\hat{A} \xrightarrow{|a_j\rangle} D$, and let $U_{j\alpha} = \langle a_j|\alpha\rangle$.

Proposition 16 (Similarity transformation).

$$\hat{A} \xrightarrow{|\alpha\rangle} A^\alpha = U^\dagger D U. \quad (51)$$

Proof. Insert the identity twice:

$$A_{\alpha\beta}^\alpha = \langle \alpha|\hat{A}|\beta\rangle = \sum_{i,j} \langle \alpha|a_i\rangle \langle a_i|\hat{A}|a_j\rangle \langle a_j|\beta\rangle = \sum_{i,j} (U^\dagger)_{\alpha i} D_{ij} U_{j\beta} = (U^\dagger D U)_{\alpha\beta}.$$

□

Result 15 (Expectation value in any basis).

$$\langle A \rangle = \langle \psi|\hat{A}|\psi\rangle = \psi^{j\dagger} D \psi^j = \psi^{\alpha\dagger} [U^\dagger D U] \psi^\alpha. \quad (52)$$

Either change the state to the eigenbasis ($\psi^j = U\psi^\alpha$) or change the operator to the state's basis ($A^\alpha = U^\dagger D U$). Both give the same number.

Problem 10 ($\hat{S}_z$ in the S_x basis). Using $U = R(\frac{\pi}{2}\hat{y})$, find the matrix of $\hat{S}_z$ in the S_x basis, and use it to redo the problem “Expectation value from the S_x basis” without changing the state’s basis.

A Appendix

B Solutions

Solution 1 (The exponential as a limit). Take the logarithm of the left side, which turns the power into a product:

$$\ln \left(1 + \frac{x}{N}\right)^N = N \ln \left(1 + \frac{x}{N}\right).$$

The Taylor series of the logarithm about $u = 0$ is

$$\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots,$$

so with $u = x/N$, which is small once N is large,

$$N \ln \left(1 + \frac{x}{N}\right) = N \left(\frac{x}{N} - \frac{x^2}{2N^2} + \frac{x^3}{3N^3} - \cdots \right) = x - \frac{x^2}{2N} + \frac{x^3}{3N^2} - \cdots.$$

Every term after the first carries a factor $1/N$ or smaller, so they all vanish as $N \rightarrow \infty$ and the logarithm tends to x . Exponentiating,

$$\lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N = e^x.$$

Solution 2 (Other bases). The same state has different columns in different bases:

$$|+\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | +\hat{z} \rangle \\ \langle -\hat{x} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_y} \begin{pmatrix} \langle +\hat{y} | +\hat{z} \rangle \\ \langle -\hat{y} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

Solution 3 (A probability by matrix mechanics). Write $|\psi\rangle$ as a column and $|+\hat{x}\rangle$ as the conjugate row:

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2i \end{pmatrix}, \quad \langle +\hat{x} | \xrightarrow{S_z} \frac{1}{\sqrt{2}} (1 \quad 1).$$

Then

$$\langle +\hat{x} | \psi \rangle = \frac{1}{\sqrt{10}} (1 \quad 1) \begin{pmatrix} 1 \\ 2i \end{pmatrix} = \frac{1+2i}{\sqrt{10}}, \quad P(S_x = +\frac{\hbar}{2}) = \frac{|1+2i|^2}{10} = \frac{5}{10} = \frac{1}{2}.$$

Take the modulus of the whole sum, not of each term separately.

Solution 4 (The matrix of $\hat{S}_z$). Here every overlap is 0 or 1 by orthonormality, so only the diagonal survives:

$$\langle +\hat{z} | \hat{S}_z | +\hat{z} \rangle = \frac{\hbar}{2} (1 - 0) = \frac{\hbar}{2}, \quad \langle -\hat{z} | \hat{S}_z | -\hat{z} \rangle = \frac{\hbar}{2} (0 - 1) = -\frac{\hbar}{2},$$

with both off-diagonal entries zero, giving

$$\hat{S}_z \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

An operator is diagonal in its own basis, and the diagonal entries are its outcomes.

Solution 5 (The matrix of $\hat{S}_x$). From $|\pm\hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}}(1, \pm 1)^T$ the overlaps are $\langle +\hat{z} | \pm\hat{x} \rangle = \langle -\hat{z} | \pm\hat{x} \rangle = \frac{1}{\sqrt{2}}$ and $\langle -\hat{z} | -\hat{x} \rangle = -\frac{1}{\sqrt{2}}$, all real. Then

$$\begin{aligned} \langle +\hat{z} | \hat{S}_x | +\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} - \frac{1}{2} \right) = 0, & \langle +\hat{z} | \hat{S}_x | -\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{\hbar}{2}, \\ \langle -\hat{z} | \hat{S}_x | +\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{\hbar}{2}, & \langle -\hat{z} | \hat{S}_x | -\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{2} - \frac{1}{2} \right) = 0, \end{aligned}$$

where the + signs come from the single minus sign carried by $\langle -\hat{z} | -\hat{x} \rangle$ meeting the minus in front of the second projector. So

$$\hat{S}_x \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Solution 6 (The matrix of $\hat{S}_y$). Now the overlaps are complex, and the bra side is conjugated: $\langle -\hat{z} | \pm\hat{y} \rangle = \pm \frac{i}{\sqrt{2}}$ while $\langle \pm\hat{y} | -\hat{z} \rangle = \mp \frac{i}{\sqrt{2}}$, with $\langle +\hat{z} | \pm\hat{y} \rangle = \langle \pm\hat{y} | +\hat{z} \rangle = \frac{1}{\sqrt{2}}$. Then

$$\begin{aligned} \langle +\hat{z} | \hat{S}_y | -\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{1}{\sqrt{2}} \cdot \frac{-i}{\sqrt{2}} - \frac{1}{\sqrt{2}} \cdot \frac{i}{\sqrt{2}} \right) = -\frac{i\hbar}{2}, \\ \langle -\hat{z} | \hat{S}_y | +\hat{z} \rangle &= \frac{\hbar}{2} \left(\frac{i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \frac{-i}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \right) = +\frac{i\hbar}{2}, \end{aligned}$$

and the diagonal entries vanish because $i \cdot (-i) = 1$ in both terms. So

$$\hat{S}_y \xrightarrow{S_z} \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Solution 7 (Expectation value and uncertainty by matrix mechanics). First move to the S_z basis:

$$\psi \xrightarrow{S_z} \frac{1}{\sqrt{10}} \left[\begin{pmatrix} 1 \\ i \end{pmatrix} + \begin{pmatrix} 2i \\ 2 \end{pmatrix} \right] = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 + 2i \\ 2 + i \end{pmatrix}.$$

Then $\langle S_z \rangle = \psi^\dagger \hat{S}_z \psi$:

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{10} \begin{pmatrix} 1 - 2i & 2 - i \end{pmatrix} \begin{pmatrix} 1 + 2i \\ -(2 + i) \end{pmatrix} = \frac{\hbar}{2} \cdot \frac{5 - 5}{10} = 0.$$

Since $\hat{S}_z^2 = \frac{\hbar^2}{4}\mathbb{1}$, we get $\langle S_z^2 \rangle = \frac{\hbar^2}{4}\psi^\dagger\psi = \frac{\hbar^2}{4}$, and therefore

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} - 0, \quad \Delta S_z = \frac{\hbar}{2}.$$

Both S_z outcomes have probability $\frac{1}{2}$ here, so the spread is maximal.

Solution 8 ($R(\phi\hat{z})$ in the S_x basis). Apply R to the S_x basis states by expanding them in $\hat{z}$:

$$R|\pm\hat{x}\rangle = \frac{1}{\sqrt{2}} \left(e^{-i\phi/2} |+\hat{z}\rangle \pm e^{i\phi/2} |-\hat{z}\rangle \right).$$

Hitting these with $\langle\pm\hat{x}| = \frac{1}{\sqrt{2}} (\langle+\hat{z}| \pm \langle-\hat{z}|)$ gives

$$\begin{aligned} \langle+\hat{x}|R|+\hat{x}\rangle &= \langle-\hat{x}|R|-\hat{x}\rangle = \frac{1}{2} \left(e^{-i\phi/2} + e^{i\phi/2} \right) = \cos \frac{\phi}{2}, \\ \langle+\hat{x}|R|-\hat{x}\rangle &= \langle-\hat{x}|R|+\hat{x}\rangle = \frac{1}{2} \left(e^{-i\phi/2} - e^{i\phi/2} \right) = -i \sin \frac{\phi}{2}, \end{aligned}$$

using the two identities from the Taylor section. So

$$R(\phi\hat{z}) \xrightarrow{S_x} \begin{pmatrix} \cos \frac{\phi}{2} & -i \sin \frac{\phi}{2} \\ -i \sin \frac{\phi}{2} & \cos \frac{\phi}{2} \end{pmatrix}.$$

Check: at $\phi = \pi/2$, $R|+\hat{x}\rangle \xrightarrow{S_x} \frac{1}{\sqrt{2}}(1, -i)^T$, which is $|+\hat{y}\rangle$ up to a global phase, as rotating $\hat{x}$ by 90° about $\hat{z}$ should give.

Solution 9 (Expectation value from the S_x basis). The state in the S_x basis is

$$|\psi\rangle \xrightarrow{S_x} \psi^x = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i\sqrt{2} \end{pmatrix}.$$

Change to the S_z basis:

$$\psi^z = R\left(\frac{\pi}{2}\hat{y}\right)\psi^x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i\sqrt{2} \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1+i\sqrt{2} \\ 1-i\sqrt{2} \end{pmatrix}, \quad \psi^{z\dagger} = \frac{1}{\sqrt{6}} (1-i\sqrt{2} \quad 1+i\sqrt{2}).$$

Then

$$\langle S_z \rangle = \psi^{z\dagger} \hat{S}_z \psi^z = \frac{\hbar}{2} \cdot \frac{1}{6} (1-i\sqrt{2} \quad 1+i\sqrt{2}) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1+i\sqrt{2} \\ 1-i\sqrt{2} \end{pmatrix} = \frac{\hbar}{2} \left(\frac{3}{6} - \frac{3}{6} \right) = 0.$$

Both S_z outcomes have probability $\frac{1}{2}$, since the two components have equal modulus.

C Useful Links

Stern-Gerlach experiment: <https://www.youtube.com/shorts/QG7kJXxN1Kk>